MANAKULA VINAYAGAR INSTITUTE OF TECHNOLOGY

An Autonomous Institution

Affiliated to Pondicherry University, Approved by AICTE, New Delhi, Accredited by NAAC with 'A' Grade

Kalitheerthalkuppam, Puducherry- 605 107.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

INTERNAL EXAMINATION - I

ject / Code : DISCRETE MATHEMATICS/CSPC401

mester : II / IV

: CSE-B

Date:13/02/2026

Time: 8.50 am to 10.30 am

Max. Marks: 50

[Bloom's Taxonomy: K1- Remember, K2- Understand, K3-Apply, K4-Analysis]

Course Name: [illegible]	Year of Study: [illegible]	Sem: [illegible]	Sub: Discrete Math (CSPC401)	Bloom's Taxonomy
Explain sets and relations, demonstrate function properties, and apply Cantor's arguments to prove cardinality distinctions.				Apply(K3)
Apply proof techniques, mathematical induction, explain number-theoretic algorithms, and solve modular arithmetic problems.				Apply(K3)

PART – A

Answer ALL the questions **(10*2=20)**

1. Define finite and infinite sets. (CO1:K1)
2. If A = {1,2,3} and B = {a, b, c} then find AxB. (CO1:K3)
3. Define Equivalence Relation. (CO1:K1)
4. Check whether the following relation $R = \{(0,0),(0,1),(0,2)(1,0),(1,1),(1,2),(2,0),(2,2),(3,3)\}$ is an Equivalence relation on a set $\{0,1,2,3\}$. (CO1:K1)
5. If $R = \{(1,1),(1,3),(3,2),(3,4),(4,2)\}$ and $S = \{(2,1),(3,3),(3,4),(4,1)\}$ then find $R \circ S$ and $S \circ R$. (CO1:K3)
6. Define Domain, Codomain of function f. (CO1:K1)
7. Determine whether the function $f : R \to R,\ f(x) = 2x + 2$ is bijective. (CO1:K3)
8. Use Contraposition to prove: If n^2 is even, then n is even. (CO2:K3)
9. State fundamental theorem of Arithmetic. (CO2:K1)
10. Find the GCD of 1819 and 3857 using Euclidean Algorithm. (CO2:K3)

PART-B

Answer ALL the questions **(3*10 = 30)**

11. (i) If R is the relation on the set of positive integers such that $(a,b) \in R$ if $a^2 + b$ is even, then Prove that R is an equivalence relation. (6)(CO1:K3)

    (ii) Prove that the relation $\subseteq$ inclusion of set is a partial ordering on any collection of sets. (4)(CO1;K3)

12. State and Prove Cantor's diagonal argument. (CO1:K3)
13. (i) Prove that √2 is irrational. (4)(CO2: K3)

    (ii) Prove that $1^2 + 2^2 + \ldots + n^2 = n(n+1)(2n+1)/6$ for all positive integers n by using mathematical induction method. (6)(CO2:K3)

********** ALL THE BEST **********